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Motivation

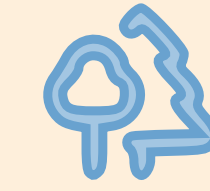


The increasing frequency of extreme weather events in Brazil highlights the vulnerability of infrastructure and human lives. Recent disasters, such as floods in Pernambuco and Minas Gerais, and the catastrophic 2024 floods in Rio Grande do Sul, demonstrates the urgent need for more accurate rainfall prediction models.

The Project: Climate Events & Disasters



Hazard: The physical threat or extreme weather event.



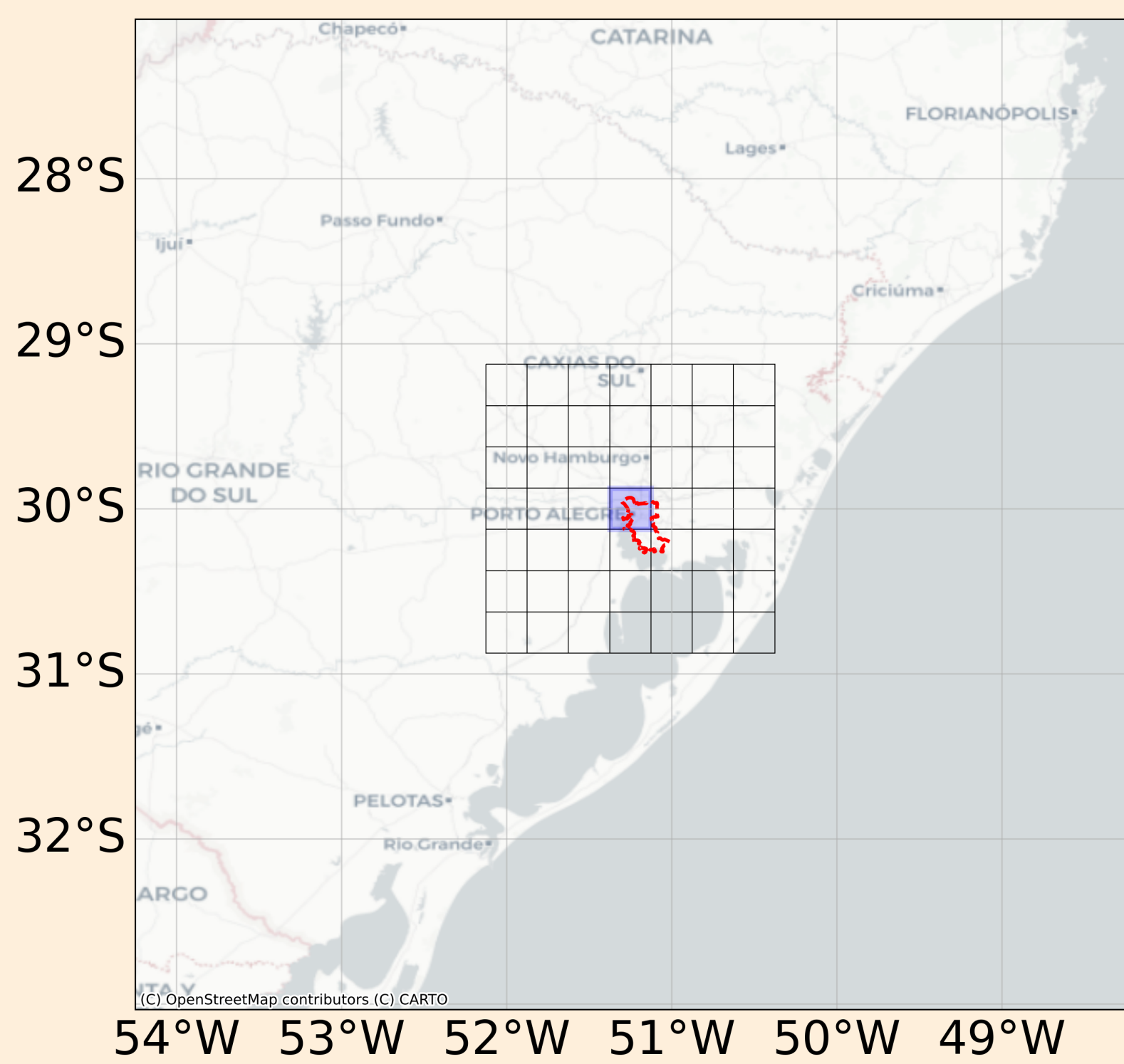
Exposure: The people and infrastructure located in harm's way.



Vulnerability: The susceptibility to damage and lack of resilience.

- Reliable forecasting is a critical first line of defense against severe weather, enabling early warnings that save lives.

Data & Methodology



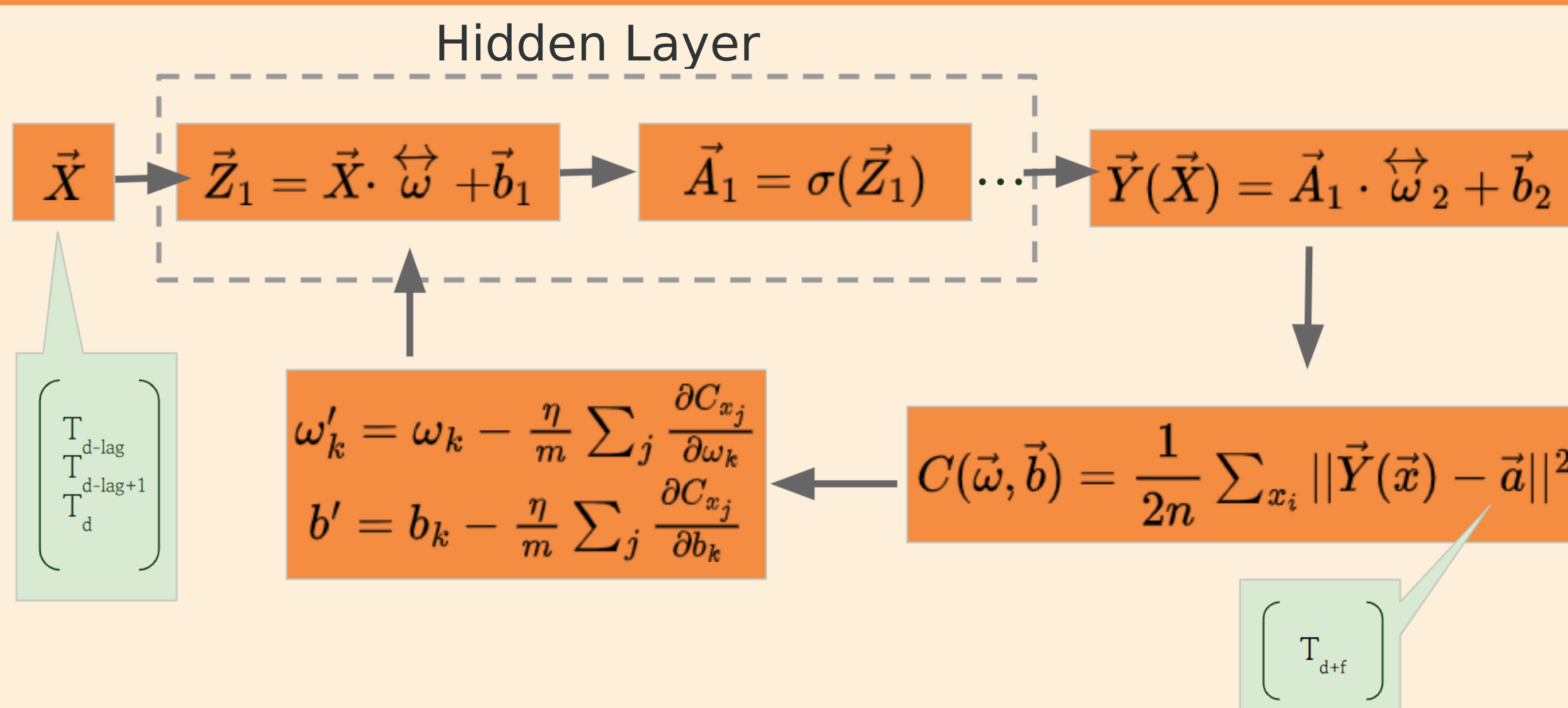
- Dataset:** Daily meteorological data was retrieved from the ERA5 Reanalysis (European Centre for Medium-Range Weather Forecasts - ECMWF). ERA5 was selected for its high global coverage and consistent physical constraints.
- Study Region:** The analysis focuses initially on Rio Grande do Sul, Southern Brazil.
- Total Precipitation (mm) and Mean Temperature (°C)** were defined as both target variables and inputs. In addition to these, the input feature space was expanded with several atmospheric predictors from the ERA5 dataset, including Mean Sea Level Pressure, Maximum Temperature, Total Column Water, Convective Available Potential Energy, and 10m Wind components (10mU, 10mV).
- Standardization:** All input features and targets were transformed using Z-score normalization:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation.

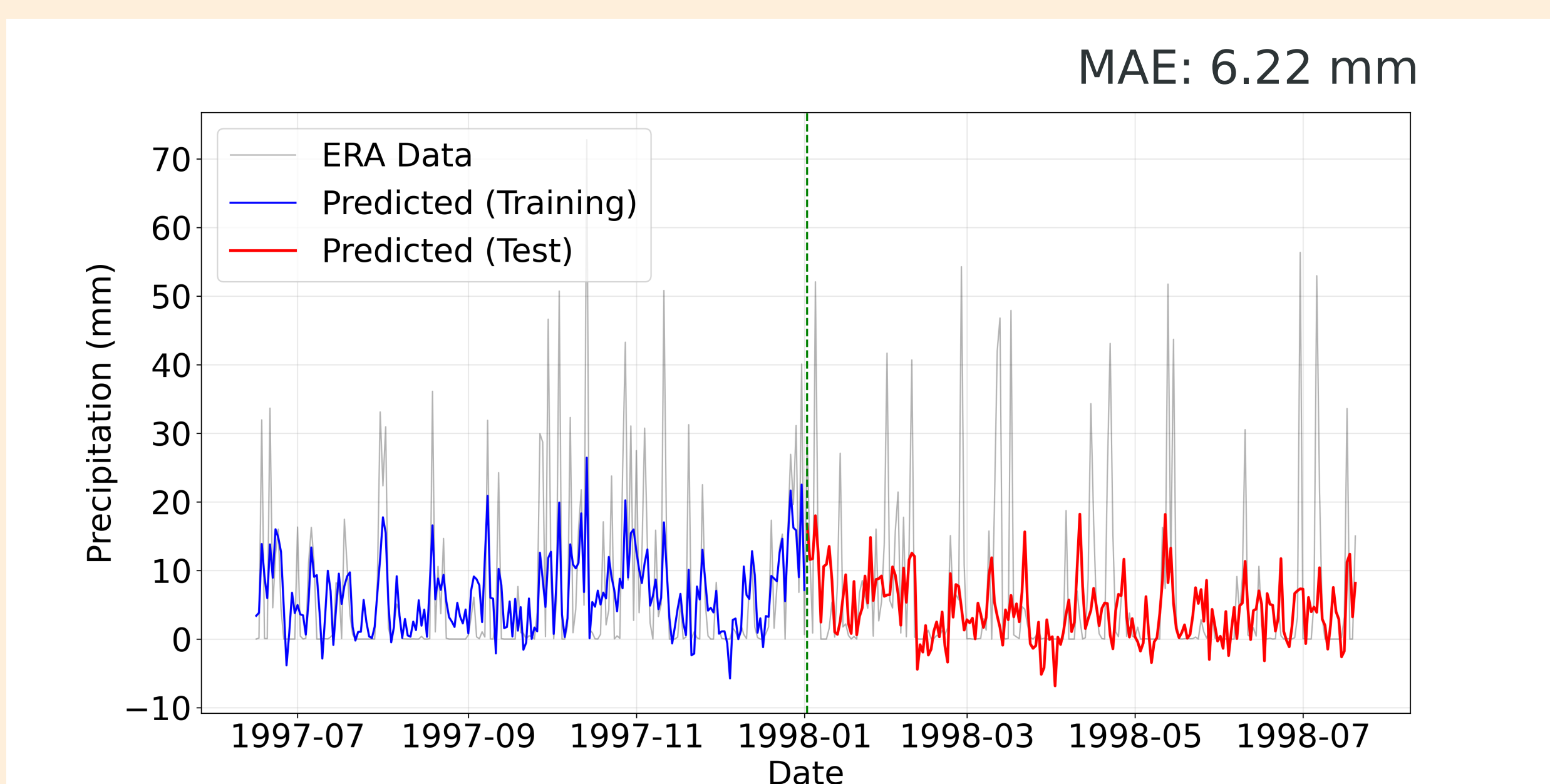
- Time-Lagged Inputs:** To provide the model with temporal context, lagged predictors were used as input features. This allows the MLP to learn the underlying persistence (or lack thereof) in the time series.

MLP - Multilayer Perceptron



- The optimization follows the Vanilla Gradient Descent algorithm. The model parameters (weights $\vec{w}$ and biases $\vec{b}$) are updated in the opposite direction of the gradient of the cost function
- Learning Rate (η):** A constant learning rate
- Backpropagation:** Gradient calculation was performed manually via the chain rule, propagating the error from the output layer back through the hidden layers to adjust the weights.
- Activation Function** Hidden layers utilize the Sigmoid and a ReLU activation function. ReLU function sets negative values to zero, mitigating the vanishing gradient problem.

Results



- Layer 1 (ReLU):** The first hidden layer utilizes the Rectified Linear Unit (ReLU) activation.
- Layer 2 (Sigmoid):** The second hidden layer employs a Sigmoid activation.
- To establish a performance benchmark, the same MLP architecture was applied to both variables temperature and precipitation. Temperature forecasting achieved high predictive skill ($MAE < 2^\circ C$) due to thermal inertia and daily persistence, contrasting with the significantly lower performance observed in the precipitation series.
- The model maintained consistent behavior across both Training and Test (Unseen) sets, though both phases exhibited the same conservative bias toward the mean.
- The minimization of Mean Squared Error via Vanilla Gradient Descent forced the model to prioritize a lower total error, resulting in a "safe" average prediction rather than capturing volatile spikes.

Conclusions

- Trend vs. Magnitude:** The MLP successfully captures the general seasonality and climatological tendencies of rainfall in Southern Brazil. However, it consistently fails to predict the exact magnitude of extreme events
- The disparity in performance highlights the distinct physical natures of the variables. While Temperature is a state-based variable with high thermal inertia (persistence), Precipitation is an intermittent flux variable with weak temporal memory, significantly increasing forecasting difficulty.

References:

- Francisco A. Rodrigues (2023) Europhysics Letters, 144(2), 22001.
- de Lima, G. R., de Oliveira Caran, R., & de Aguiar Filho, L. F. (2023) Earth Science Informatics, 16(4), 4313-4326